

FINAL ML PROJECT

# PREDICTIVE MAINTENANCE

Machine Learning Methods and Models for Predicting  
the Technical Condition of Industrial Equipment

Can sensor data identify operating conditions linked to  
machine failure?

**0.898**

PR-AUC

**83.9%**

Recall

**47 / 56**

Failures found

**14**

False alarms

# PROBLEM & SUCCESS CRITERIA

02

Predictive maintenance should detect the rare cases that matter – not just maximise accuracy.

## 01 REACTIVE

Repair after failure  
Downtime has already happened

## 02 PREVENTIVE

Service on a schedule  
Can maintain healthy equipment

## 03 PREDICTIVE

Use current condition  
Prioritise equipment for inspection

## SUCCESS CRITERIA

- Detect a high share of real failures → Recall / F2
- Evaluate rare positives correctly → PR-AUC
- Keep alerts useful → Precision
- Beat a simple baseline on a locked test set

# CONTEXT: PREDICTIVE MAINTENANCE IN KAZAKHSTAN

03

The research direction already exists; this project focuses on a rigorous, reproducible baseline.

**2024**

## Oil-well failure prediction

~4,000 wells; LR / Random Forest / Naive Bayes

SOCAR Proceedings

**2026**

## Operating-profile clustering

K-means to group oil-well operating behaviour

SOCAR Proceedings

**2026**

## Bearing condition & RUL

RF / XGBoost / LSTM + SHAP

EKTU JICS

**OUR CONTRIBUTION** leakage-safe pipeline • imbalance-aware evaluation • feature ablation • error analysis • SHAP • working API

# DATA: AI4I-STYLE PREDICTIVE MAINTENANCE

04

10,000 observations, six predictive inputs, rare failure class.

**10,000**

Rows

**6**

Input features

**0**

Missing

**3.73%**

Failure rate

## INPUTS

Type (L/M/H)

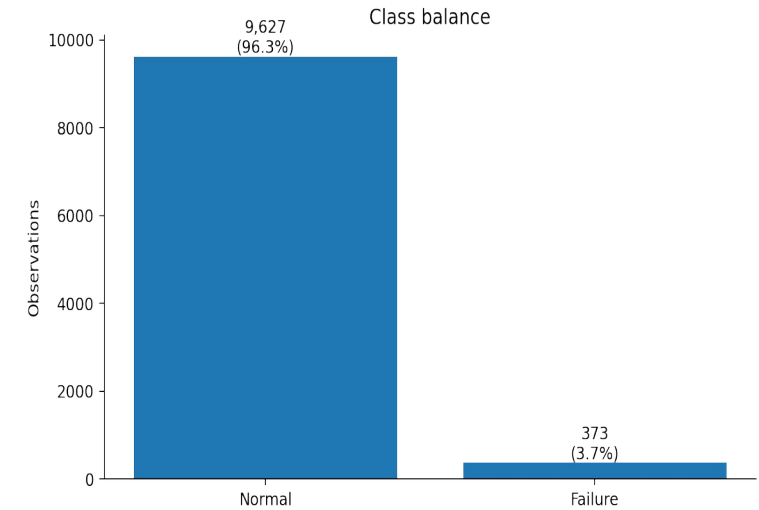
Air temperature

Process temperature

Rotational speed

Torque

Tool wear



**DATA NOTE** This run uses a bundled offline replica built from the public UCI AI4I specification. It is not the byte-identical official UCI CSV; all reported metrics in this deck refer to this replica.

# DATA PREPARATION & LEAKAGE CONTROL

05

Only information available at inference time is allowed into the model.

## STRATIFIED SPLIT

**70%**

TRAIN

7,000

**15%**

VALIDATION

1,500

**15%**

LOCKED TEST

1,500

## REMOVED FROM PREDICTIVE INPUTS

- UDI + Product ID → identifiers
- TWF / HDF / PWF / OSF / RNF → target leakage

Failure-mode labels are used only after prediction for error analysis.

## 3 ENGINEERED FEATURES

### Temperature gap

process temp - air temp

### Mechanical power

$\text{torque} \times \text{RPM} \times 2\pi/60$

### Wear × torque

tool wear × torque

# EXPERIMENT DESIGN

06

One pipeline, fair model comparison, validation-only threshold selection.

## 1 PREPROCESS

impute • scale • one-hot

## 2 BASELINE

Dummy + Logistic Regression

## 3 TREE MODELS

Random Forest + XGBoost

## 4 VALIDATE

choose threshold by F2

## 5 TEST

locked test metrics

## 6 EXPLAIN

SHAP + error analysis

**WHY F2?** A false negative means a real failure was missed. F2 weights recall more than precision, so the validation threshold is chosen to reduce missed failures.

# EXPERIMENT: FEATURE ENGINEERING & THRESHOLD

07

We tested not only algorithms, but also whether the representation and decision rule improve the result.

## FEATURE ABLATION – XGBoost

| Input set             | PR-AUC       | Recall       | F1           |
|-----------------------|--------------|--------------|--------------|
| Raw 6 features        | 0.798        | 0.804        | 0.714        |
| <b>+ 3 engineered</b> | <b>0.895</b> | <b>0.821</b> | <b>0.829</b> |

**PR-AUC +0.097**

Engineered interactions make the failure rules easier for tree models to learn.

## DECISION THRESHOLD

**0.375**

selected on validation by F2

Default **0.50**

Recall 82.1% • F2 0.816

**Selected 0.375**

Recall 83.9% • F2 0.825

The threshold changes the operating point; it does not retrain the model.

# HELD-OUT TEST RESULTS

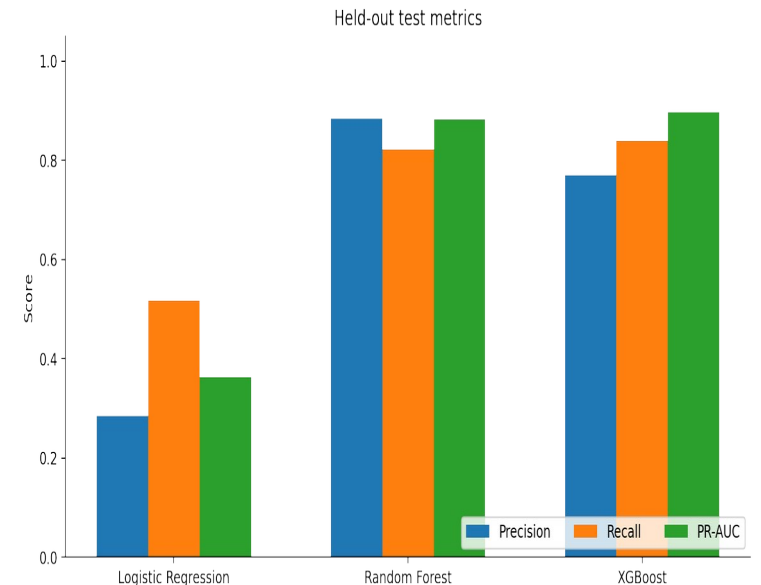
08

Tree ensembles strongly outperform the linear baseline on the rare failure class.

## MODEL COMPARISON

| Model               | Prec.        | Recall       | F1           | F2           | PR-AUC       |
|---------------------|--------------|--------------|--------------|--------------|--------------|
| Dummy               | 0.000        | 0.000        | 0.000        | 0.000        | 0.037        |
| Logistic Regression | 0.284        | 0.518        | 0.367        | 0.445        | 0.362        |
| Random Forest       | 0.885        | 0.821        | 0.852        | 0.833        | 0.883        |
| <b>XGBoost</b>      | <b>0.770</b> | <b>0.839</b> | <b>0.803</b> | <b>0.825</b> | <b>0.898</b> |

**Dummy accuracy = 96.3% — but it detects 0 failures.**



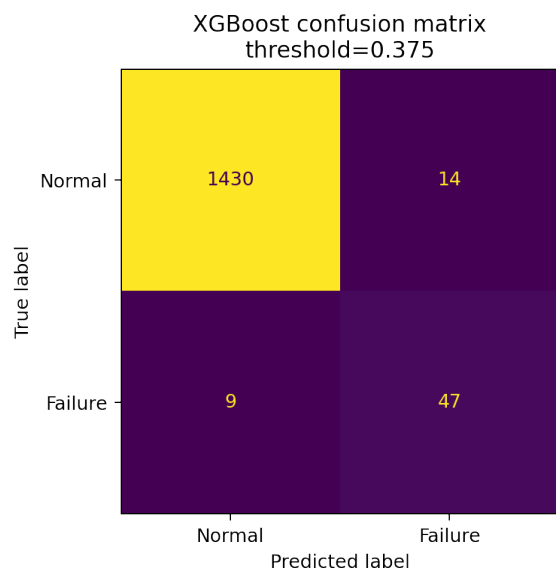
Best PR-AUC

**XGBoost 0.898**

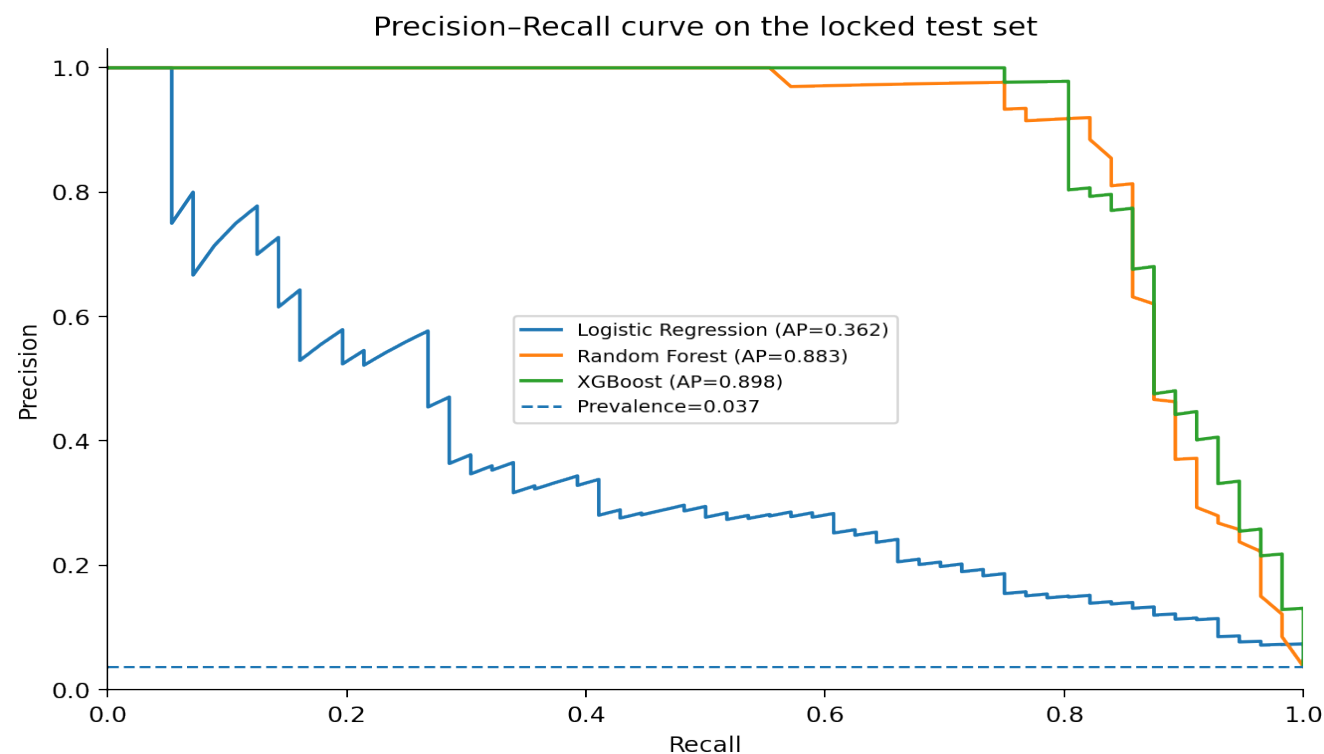
# FINAL XGBOOST ON THE LOCKED TEST SET

09

The model finds 47 of 56 failures while producing 14 false alarms.



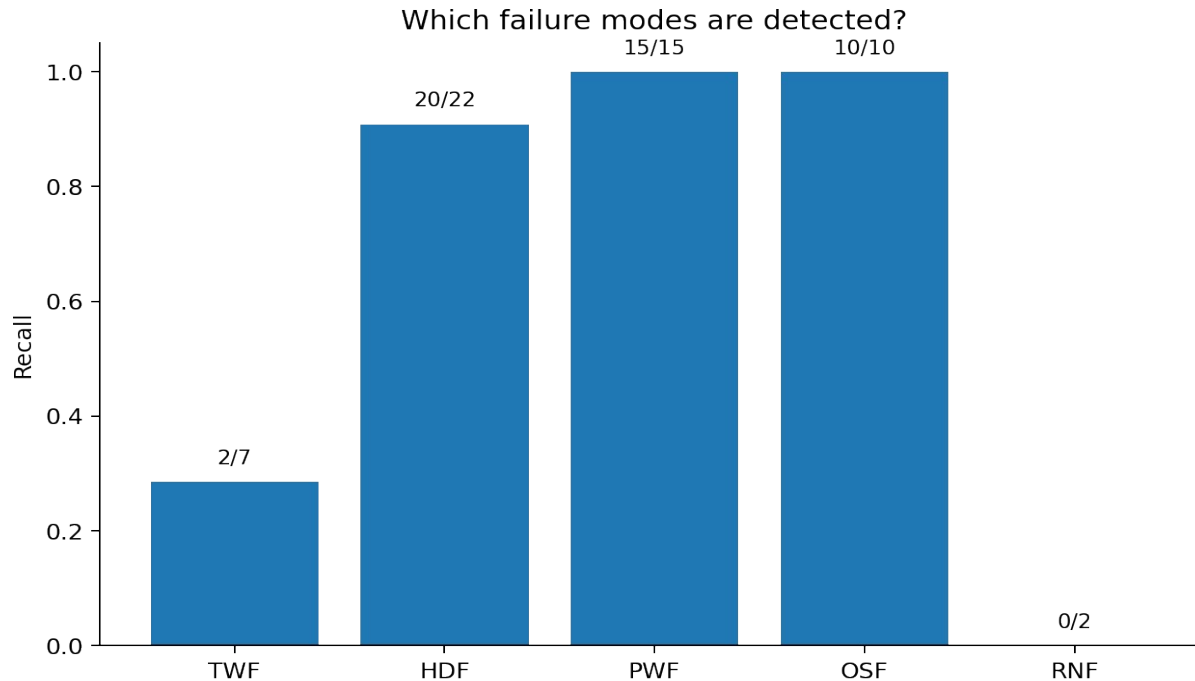
**47 TP**   **9 FN**   **14 FP**



# ERROR ANALYSIS: NOT EVERY FAILURE IS PREDICTABLE

10

Hidden failure-mode labels were inspected only after prediction to understand the nine false negatives.



## WHAT THE ERRORS TELL US

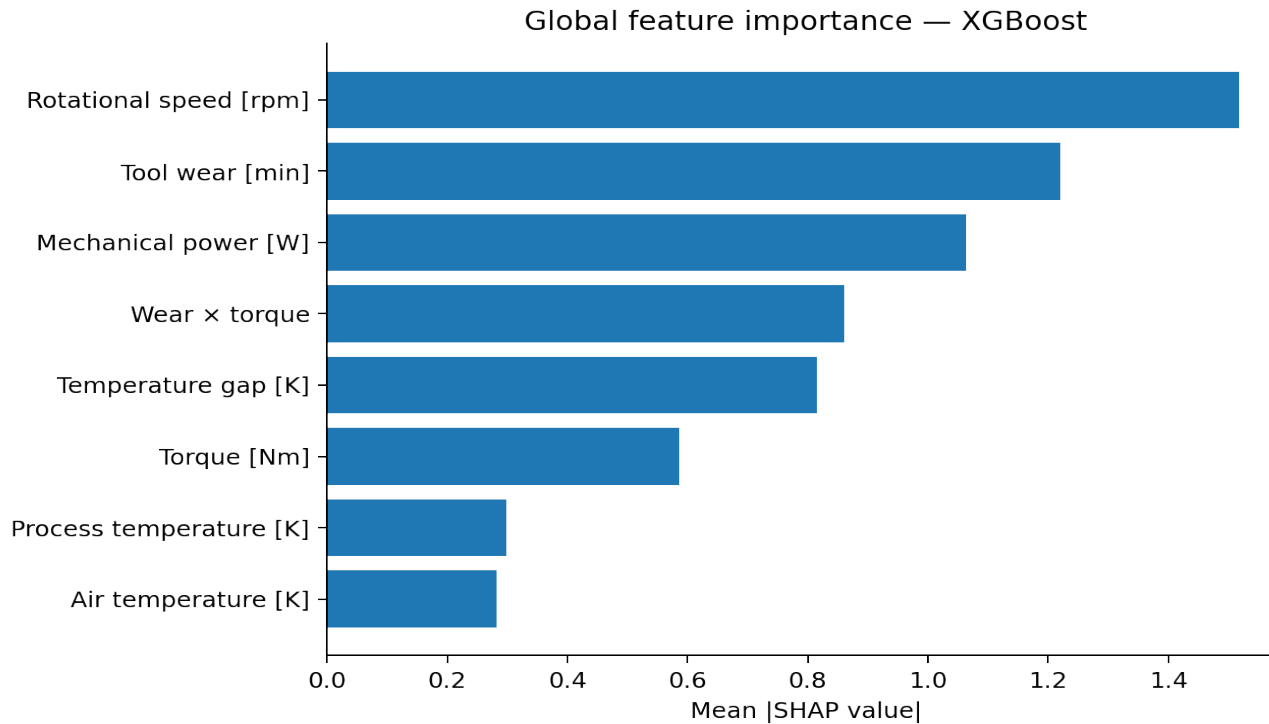
- PWF: 15 / 15 detected
- OSF: 10 / 10 detected
- HDF: 20 / 22 detected
- TWF: only 2 / 7 detected
- RNF: 0 / 2 detected

Some TWF/RNF events are random by construction. Missing information cannot be recovered by a stronger model.

# EXPLAINABILITY WITH SHAP

11

SHAP explains model behaviour; it does not prove physical causality.



## TOP DRIVERS

1 Rotational speed

2 Tool wear

3 Mechanical power

4 Wear x torque

5 Temperature gap

### Use case

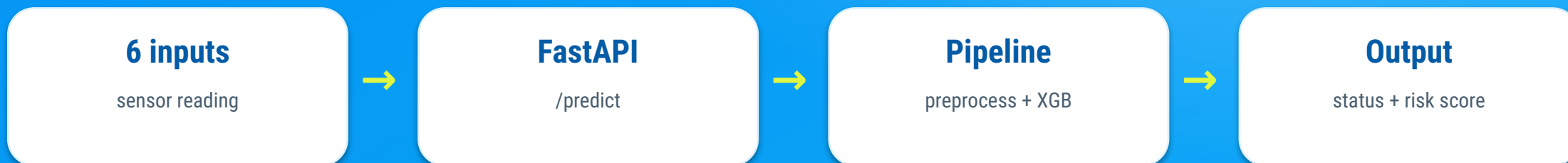
Explain why a reading received a high model score and support an engineer's review.

# WORKING PROTOTYPE & ENGINEERING QUALITY

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The same preprocessing and model artifact are used in training and inference.

## DEMO FLOW



## REPRODUCIBILITY

- Fixed random seed
- sklearn Pipeline
- requirements.txt
- Dockerfile + smoke test

## RESPONSIBLE AI

- risk score  $\neq$  calibrated probability
- human-in-the-loop
- no automatic machine shutdown
- real telemetry + drift monitoring required

# CONCLUSION

## WHAT WE ACHIEVED

- Leakage-safe, reproducible ML pipeline
- Random Forest / XGBoost strongly beat the linear baseline
- XGBoost: PR-AUC 0.898, recall 83.9% on locked test
- Feature engineering produced a measurable gain
- Error analysis exposed predictable and unpredictable failure modes
- Working FastAPI prototype + SHAP explanations

## NEXT STAGE

- Official / real industrial data
- Time-series telemetry
- Remaining Useful Life
- Cost-sensitive decisions
- MLOps + drift monitoring

## KEY REFERENCES

UCI AI4I 2020 Predictive Maintenance Dataset, DOI 10.24432/C5HS5C • Matzka (2020), AI4I • Aktaukenov et al. (2024, 2026), SOCAR Proceedings • Zharlykassova et al. (2026), EKTU JICS

THANK YOU • QUESTIONS?